

TaxaFetch

Taxonomic assignments are more accurate when they consider which species are plausible at the sampling location. Without this context, classifiers frequently assign detections to ecologically implausible taxa. TaxaFetch compiles species occurrence records from multiple sources – GBIF, DataONE, BioTIME, and published literature – and combines them into a standardized format for downstream habitat assignment (TaxaHabitat) and prior estimation (TaxaExpect).

Data Sources

Source	Function	What it provides
GBIF	<code>fetch_gbif_occurrences()</code>	Global occurrence records via download API
DataONE	<code>fetch_dataone_occurrences()</code>	Ecological datasets from DataONE repositories
BioTIME	<code>read_biotime_study()</code>	Time-series biodiversity data
Literature	<code>search_literature()</code>	OpenAlex scholarly search + PDF download
PDFs	<code>extract_pdf_text()</code>	Extract occurrence data from published PDFs

Installation

```
# Requires TaxaTools (foundation package)
devtools::install("path/to/TaxaTools")
devtools::install("path/to/TaxaFetch")
```

Quick Start

```
library(TaxaFetch)

# 1. Define a bounding box (Southern California coast)
bbox <- make_bbox_wkt(lat_min = 33.5, lat_max = 34.5,
                      lon_min = -120.0, lon_max = -118.5)

# 2. Get GBIF taxon keys from verified names
keys <- get_keys_from_context(
  c("Fundulus parvipinnis", "Atherinops affinis"),
  backbone_id = 11 # GBIF backbone
)

# 3. Fetch GBIF occurrences
gbif_data <- fetch_gbif_occurrences(keys = keys$usageKey, geometry = bbox)

# 4. Filter to high-quality records
filtered <- filter_gbif_quality(gbif_data, max_coord_uncertainty = 500)
```

```
# 5. Stack multiple sources into one table
all_data <- stack_occurrences(gbif = filtered)
```

Key Functions

GBIF pipeline: - `make_bbox_wkt()` – create WKT bounding box for spatial queries - `get_keys_from_context()` – resolve taxon names to GBIF usage keys - `fetch_gbif_occurrences()` – download occurrence records - `filter_gbif_quality()` – remove low-quality records

DataONE pipeline: - `search_dataone()` – find datasets by taxon and location - `harvest_dataone_catalog()` – build a dataset catalog with LLM screening - `fetch_dataone_occurrences()` – download and standardize occurrence tables

Literature + PDF pipeline: - `search_literature()` – OpenAlex scholarly search - `download_literature_pdfs()` – batch PDF download - `extract_pdf_text()` – section-aware text extraction - `screen_pdf_structure()` – LLM-based relevance screening - `build_pdf_extract_prompt()` / `parse_pdf_extract_response()` – LLM data extraction

Combining sources: - `stack_occurrences()` – row-bind sources into a single standardized table - `report_fetch()` – summarize data acquisition for reports

API Keys

TaxaFetch uses GBIF credentials for occurrence downloads and OpenAlex for literature search. See the TaxaTools API Setup vignette for configuration.

Vignettes

- Data Acquisition – end-to-end guide

Part of TaxaID

TaxaFetch is the data acquisition layer in the TaxaID ecosystem. Its output feeds into TaxaHabitat (habitat assignment) and TaxaExpect (prior estimation).

Ecosystem: TaxaTools -> **TaxaFetch** -> TaxaHabitat -> TaxaExpect -> TaxaAssign

See the TaxaID README for ecosystem overview and installation instructions.

Citation

Lafferty, K.D., 2026, TaxaID – A modular R ecosystem for Bayesian taxonomic assignment: U.S. Geological Survey software release, <https://doi.org/10.5066/xxxxxx>.

Software Requirements

- R ($\geq 4.1.0$)
- TaxaTools (foundation package, installed first)
- rgbif (for GBIF occurrence queries)
- An LLM API key is needed for literature screening and PDF extraction

All dependencies are declared in the DESCRIPTION file and installed automatically.
Developed with Claude Code (Anthropic).